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$$\sum_{n=1}^{\infty} \frac{n!}{7^n}$$

criterium d'Alembert

$$\lim_{n \rightarrow +\infty} \frac{(n+1)! \cdot 7^n}{7^{n+1} \cdot n!} = \lim_{n \rightarrow +\infty} \frac{(n+1) \cdot \cancel{n!} \cdot \cancel{7^n}}{\cancel{7^n} \cdot 7 \cdot \cancel{n!}} = +\infty > 1$$

$\Rightarrow$ reeks is divergent

References